Biophysics (Bio215)

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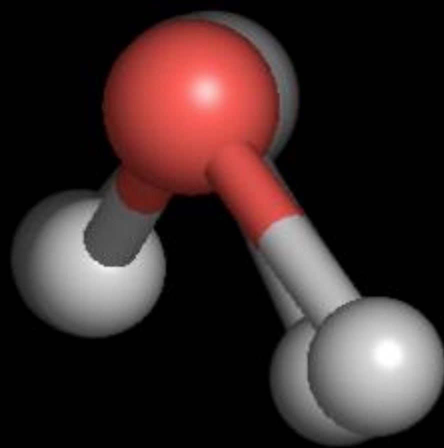
Molecular interaction, the Boltzmann Distribution & Entropy

Outline for today

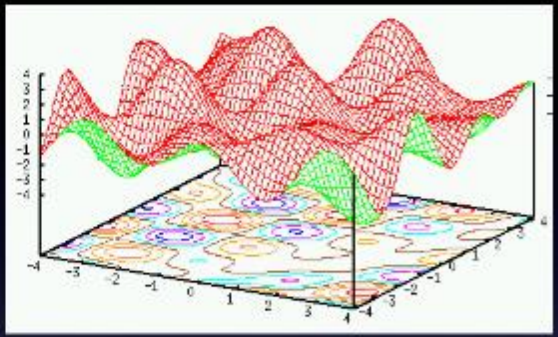
- Boltzmann
- Introduction to entropy and free energy
- Electrostatics and charges in biological materials

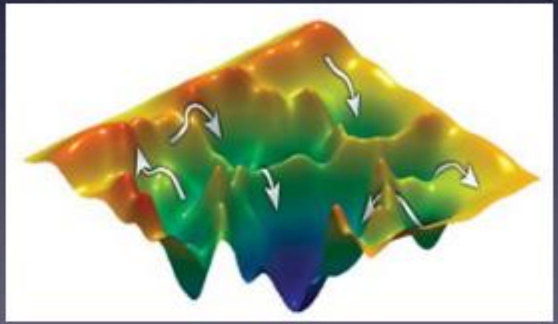
8 fs

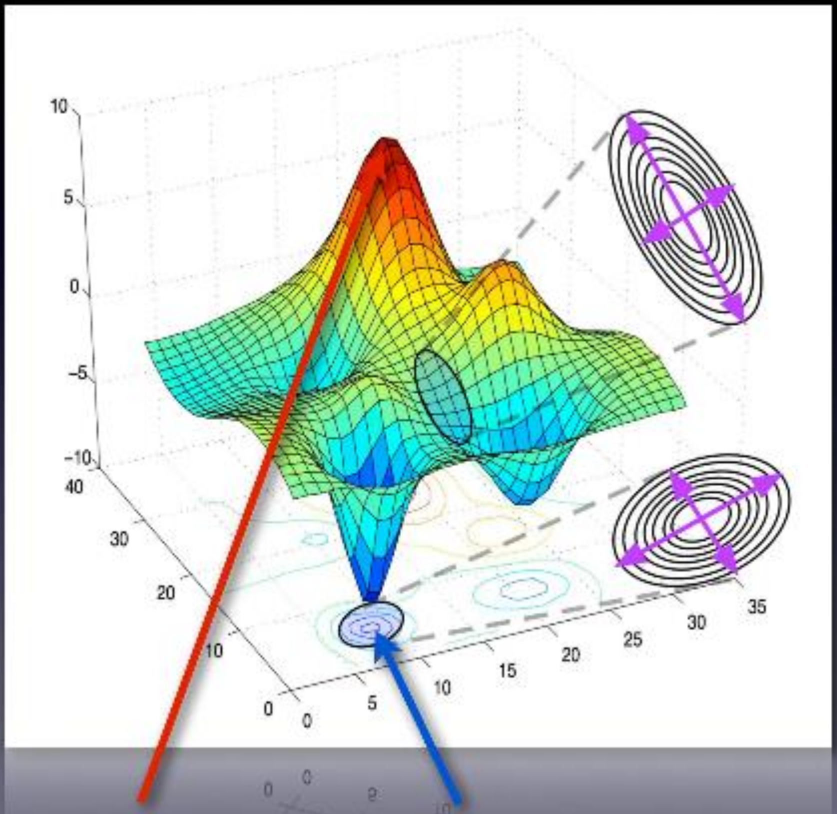


$$
\begin{aligned}
V(r) = & \sum_{bonds} \frac{1}{2} k^b_{ij} \left(r_{ij} - r^0_{ij} \right)^2 \\
& + \sum_{angles} \frac{1}{2} k^\theta_{ijk} \left(\theta_{ijk} - \theta^0_{ijk} \right)^2 \\
& + \sum_{torsions} \left\{ \sum_n k_\theta \left[1 + \cos\left(n\phi - \phi_0 \right) \right] \right\} \\
& + \sum_{impropers} k_\xi \left(\xi_{ijkl} - \xi^0_{ijkl} \right) \\
& + \sum_{i,j} \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}} \\
& + \sum_{i,j} \left[\frac{C_{12}}{r^{12}_{ij}} - \frac{C_6}{r^6_{ij}} \right]
\end{aligned}
$$

Energy landscapes

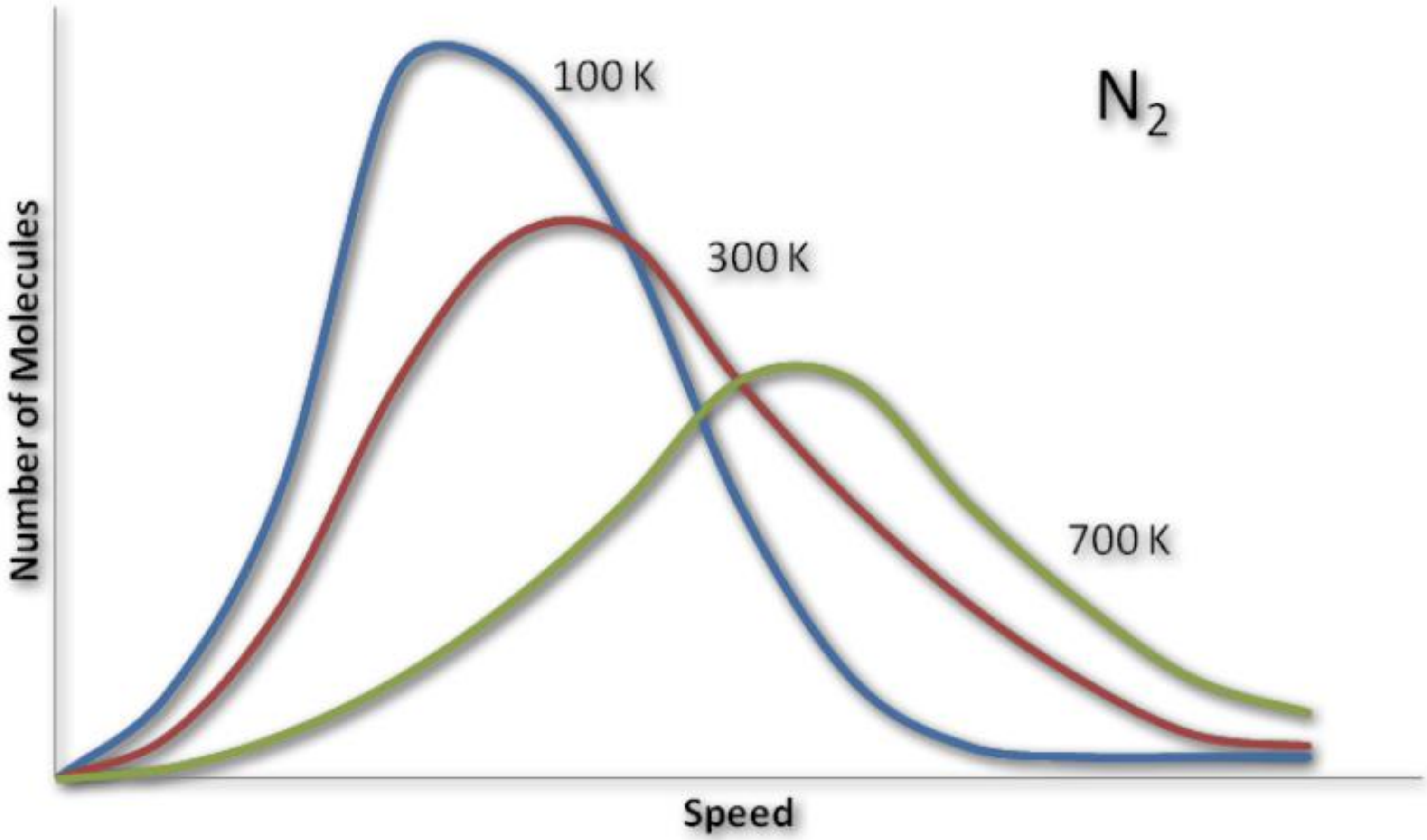




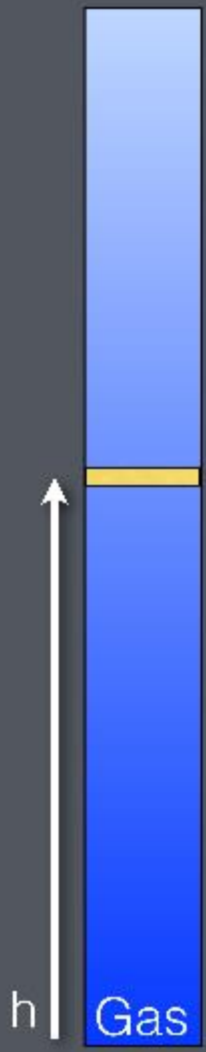


Bad?

Good?


100 K
N_2
300 K
700 K
Number of Molecules
Speed

Deriving the Boltzmann distribution for one case



low density,
high potential energy

How many molecules
do we have here?

This is a function of h,
or similarly: the potential energy

high density,
low potential energy

Gravity ↓ ↑ Pressure

Clapeyron's gas law:

$$pV = NkT$$

Or, if you are a chemist:

$$pV = NRT$$

$$n = \frac{N}{V}$$

We don't have a specific volume

$$p = nkT$$

Pressure as function of n/V

$$\frac{dp}{dh} = \frac{dn}{dh}kT$$

How does p vary with height?
(as the number of particles do)

Pressure was acting up - gravity is acting down

$$E(h) = mgh$$ Potential energy as function of h

If we go up by dh, the weight of the gas pressing down decreases by: $(mgn)dh$

Since we also had (last slide) $\frac{dp}{dh} = \frac{dn}{dh}kT$

Those changes must be equal: $$\frac{dn}{dh}kT = -mgn$$

Which we can also write $$\frac{dn}{dh} = -\frac{mg}{kT}n$$

This is a differential equation (n on right-hand side too)

Use a little trick from mathematics - we know that

$$\frac{d\left[\ln n\right]}{dh} = \frac{1}{n}\frac{dn}{dh}$$

So we can write

$$\frac{d\left[\ln n\right]}{dh} = -\frac{mg}{kT}$$

Integrate the constant right side, and take the exponent:

$$n \propto \exp^{-\frac{mgh}{kT}} = \exp^{-\frac{\Delta E(h)}{kT}}$$

Q.E.D.

The Boltzmann distribution

$$\rho \propto e^{-\Delta E/kT}$$

What does Boltzmann mean?

Probability of being in state A, with energy E_A:

$$p(E_A) = C \exp^{-\frac{E_A}{kT}}$$

Probability of being in state B, with energy E_B:

$$p(E_B) = C \exp^{-\frac{E_B}{kT}}$$

Relative probability:
(note how C cancels)

$$\frac{p(E_A)}{p(E_B)} = \frac{\exp^{-\frac{E_A}{kT}}}{\exp^{-\frac{E_B}{kT}}} = \exp^{-\frac{\Delta E_{A-B}}{kT}}$$

Lower-energy states will be more populated

Which shape in our example would be best energy-wise?

The volume seems to matter?

Counting states (volume)

Let's allow state A to have volume V_A, and similar for B

The number of states is proportional to volume

Then… the probability is also proportional to volume

So, the probability of finding particles in volume A compared to volume B becomes

$$\frac{pV_A}{pV_B} = \frac{V_A \exp^{-\frac{E_A}{kT}}}{V_B \exp^{-\frac{E_B}{kT}}}$$

Hm, makes sense, but looks complicated. Let's try something…

Use the trick to write $V = \exp\left(\ln(V)\right)$

$$\frac{pV_A}{pV_B} = \frac{\exp^{-\frac{E_A}{kT}+\ln(V_A)}}{\exp^{-\frac{E_B}{kT}+\ln(V_B)}} = \frac{\exp^{-\frac{E_A - Tk\ln(V_A)}{kT}}}{\exp^{-\frac{E_B - Tk\ln(V_B)}{kT}}}$$

This looks just like a Boltzmann distribution?
But now it says E-T*k ln(V) instead of E

We call this the "free energy" rather than energy, since it typically describes how much energy is available to perform work in a system.
But what is the "volume"? Can we make it simpler?

Entropy & Free Energy

When physicists can't simplify more, we define things!

Let's call this extra part "entropy": $S = k \ln V$

(Note how we leave out the temperature)

And we can write our free energy as

$$F = E - TS$$

Describes energy of each state

Describes #states with this E

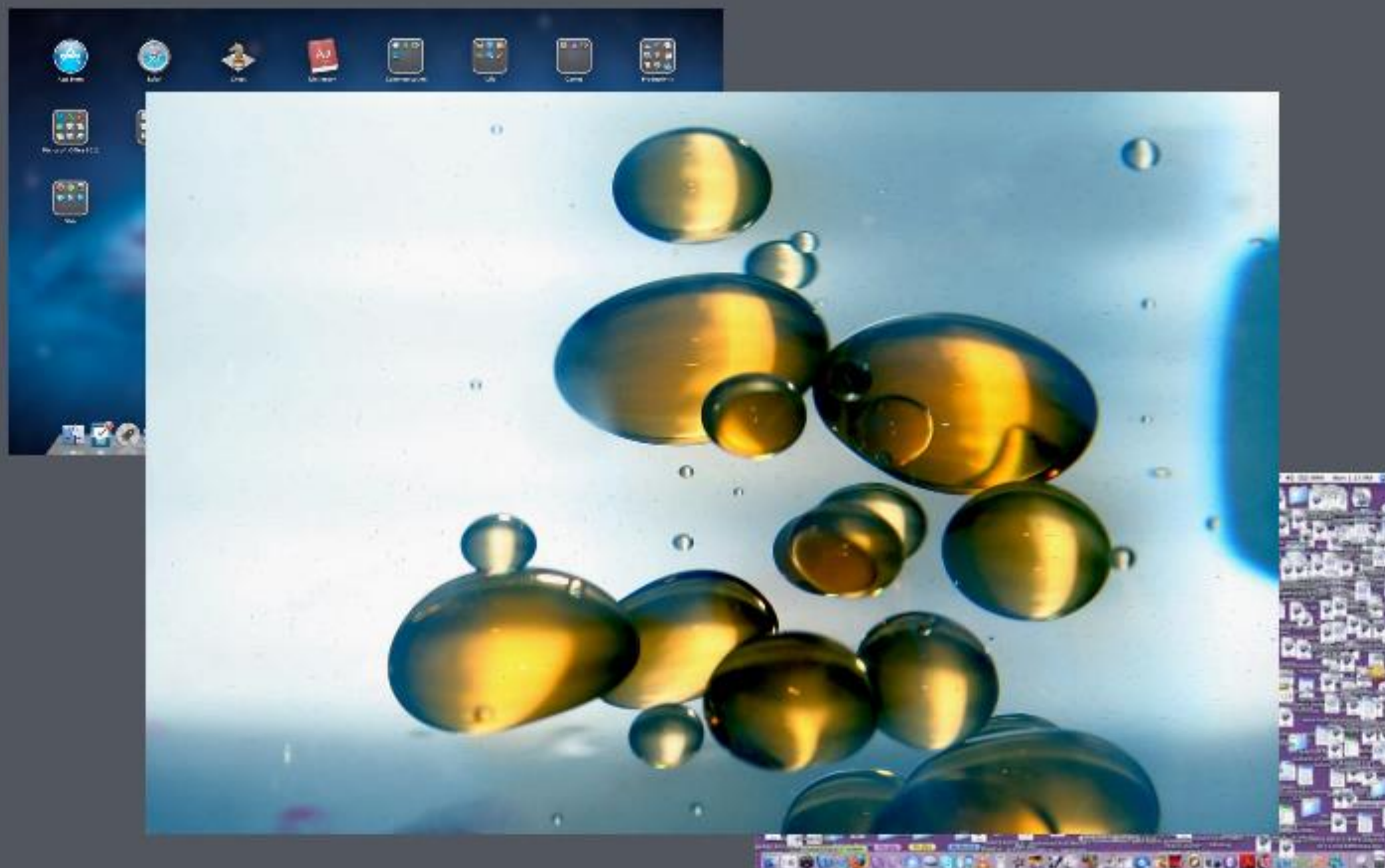
$$\frac{p_A}{p_B} = \exp^{-\frac{\Delta F}{kT}}$$

How many states does this correspond to?

How many similar states are there?

How many states does this correspond to?

How many similar states are there?



Laws of Thermodynamics

1. Energy can neither be created nor destroyed, only change form

2. The entropy of an isolated system not in equilibrium will increase over time, and approach a maximum when it reaches equilibrium

3. As temperature approaches absolute zero (0 K), the entropy of any system approaches a minimum

Helmholtz vs. Gibbs

Heat

$$F = E - TS$$

Helmholtz free energy

Heat

$$G = E + pV - TS$$

$$G = H - TS$$

Gibbs free energy

Thermodynamic temperature

What is "temperature"?

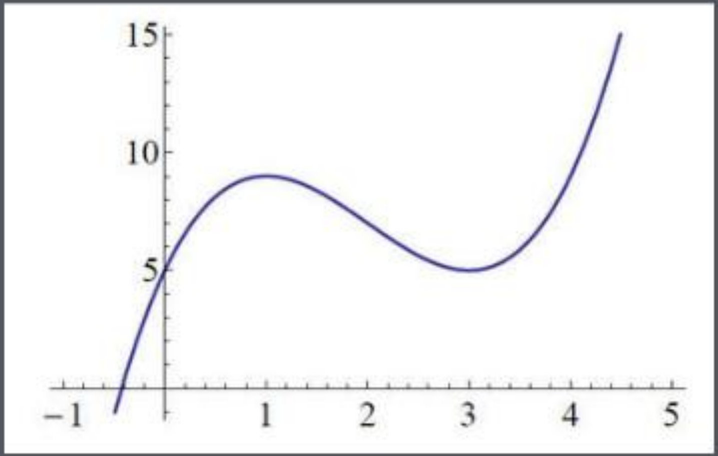
For all we know, it is just a constant in the equations.

Start from F = E-TS and look at small changes (dF):

$$F = E - TS$$

$$F + dF = F + dE - TdS - SdT$$

At equilibrium we have constant T, and we should have a local minimum in F (dF=0)



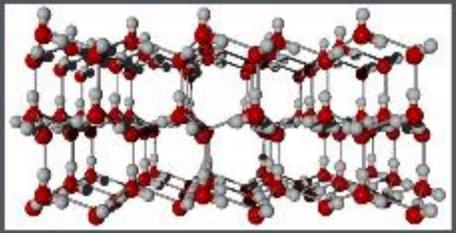
$$dF = dE - TdS = 0$$

$$T = \frac{dE}{dS}$$

Thermodynamic *definition* of temperature - derived from E,S!

Phase transitions explained

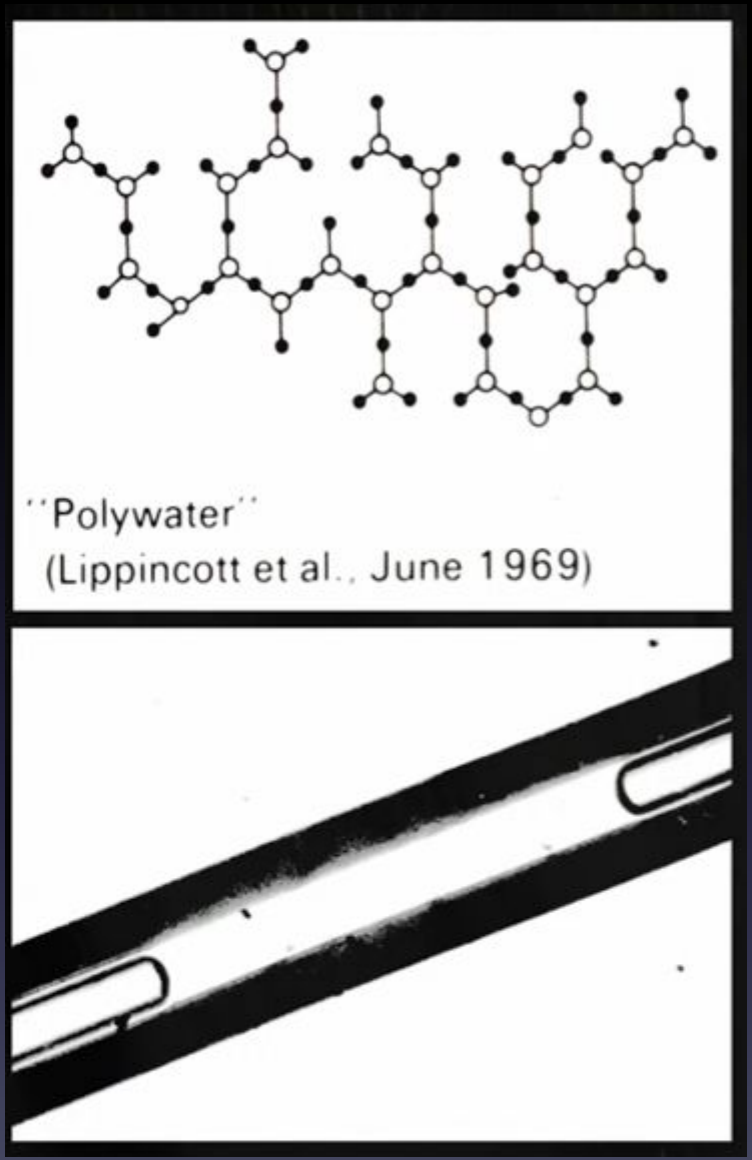
F = E - TS, and the system wants to stay at lowest F



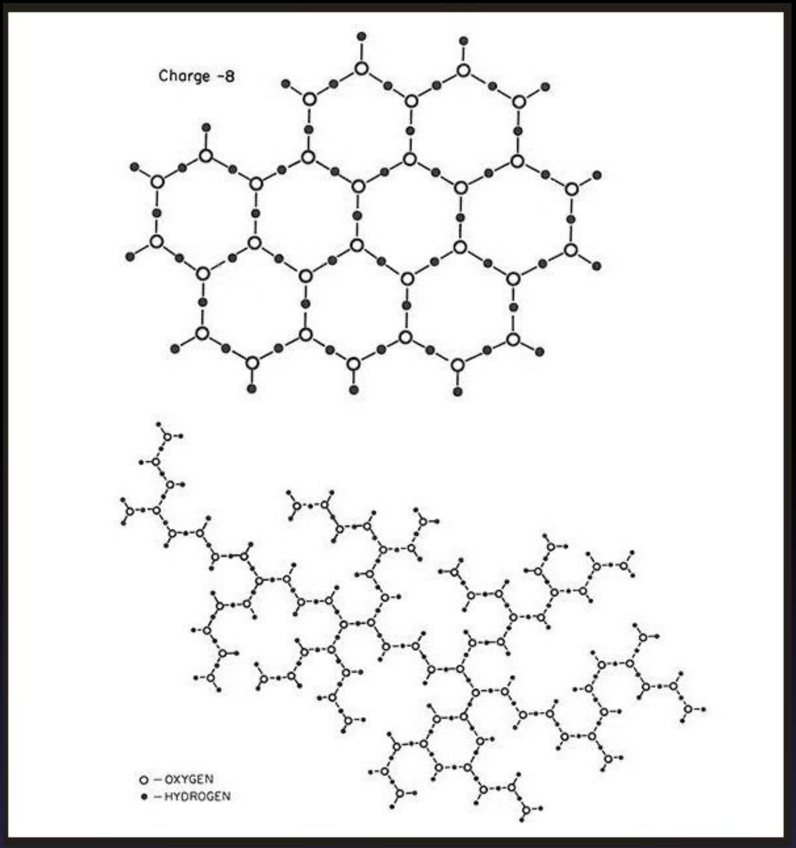
low E
low low S

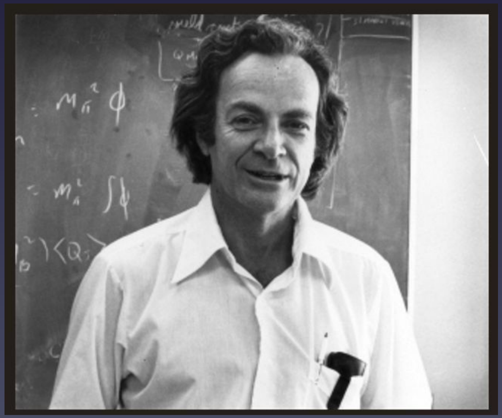
high E
medium S

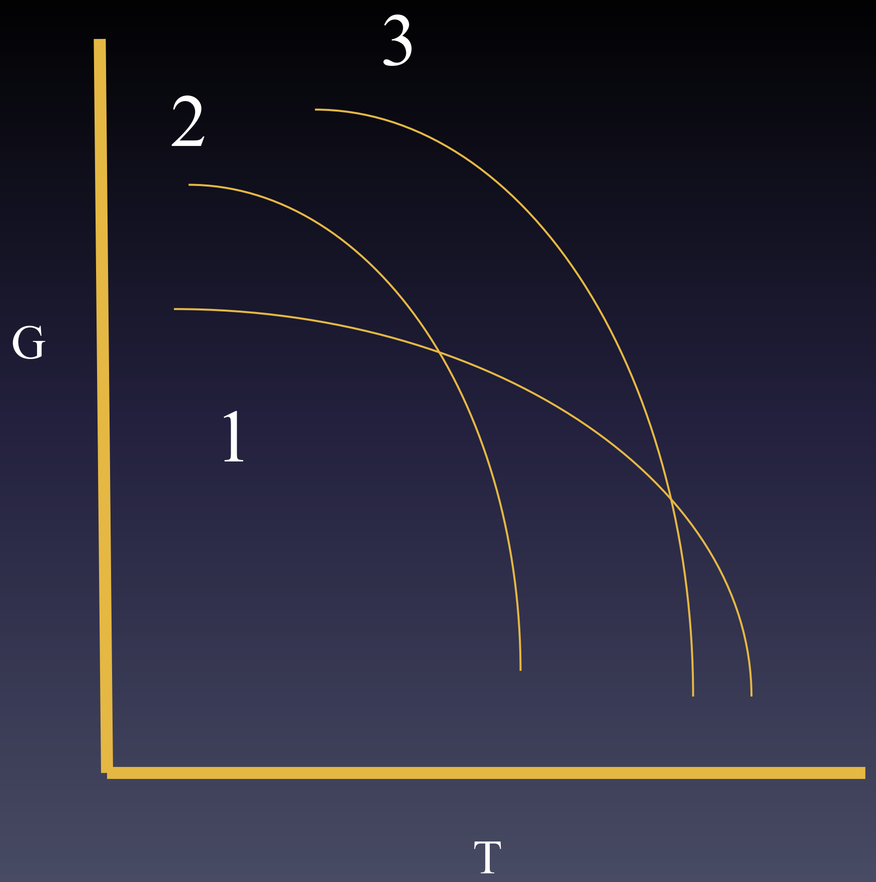
At low T, first term of F (E) dominates.
When T is higher, second term of F (TS) dominates

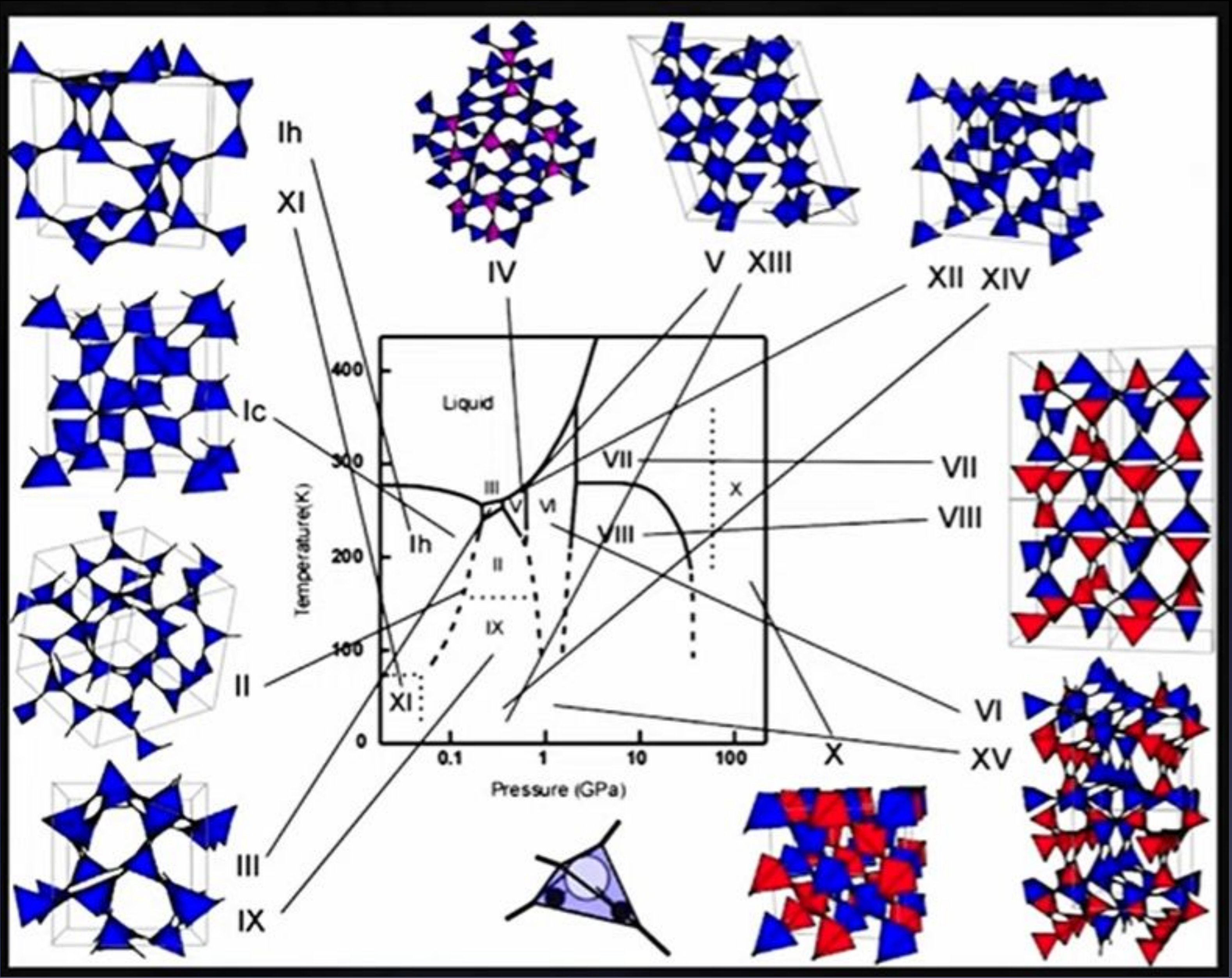


Hypothesis: It was believed that water molecules could link into polymer-like chains, creating a substance with distinct properties, such as a boiling point above 100°C and freezing below 0°C


Charge -8
O – OXYGEN
• – HYDROGEN

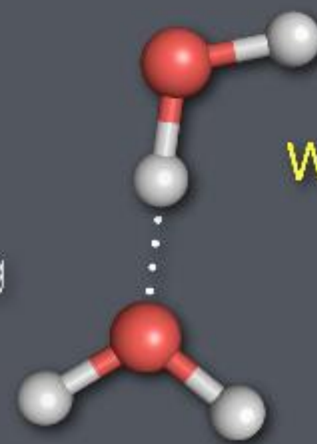



3
2
G
1
T


Ih
XI
IV
V XIII
XII XIV
Ic
Liquid
VII
VIII
X
III
V
VI
Ih
II
IX
XI
Temperature(K)
400
300
200
100
0
0.1
1
10
100
Pressure (GPa)
II
III
IX
VII
VIII
VI
X
XV

Vacuo

Water



Gain: Energy of 1 h-bond ($\Delta E_H<0$)

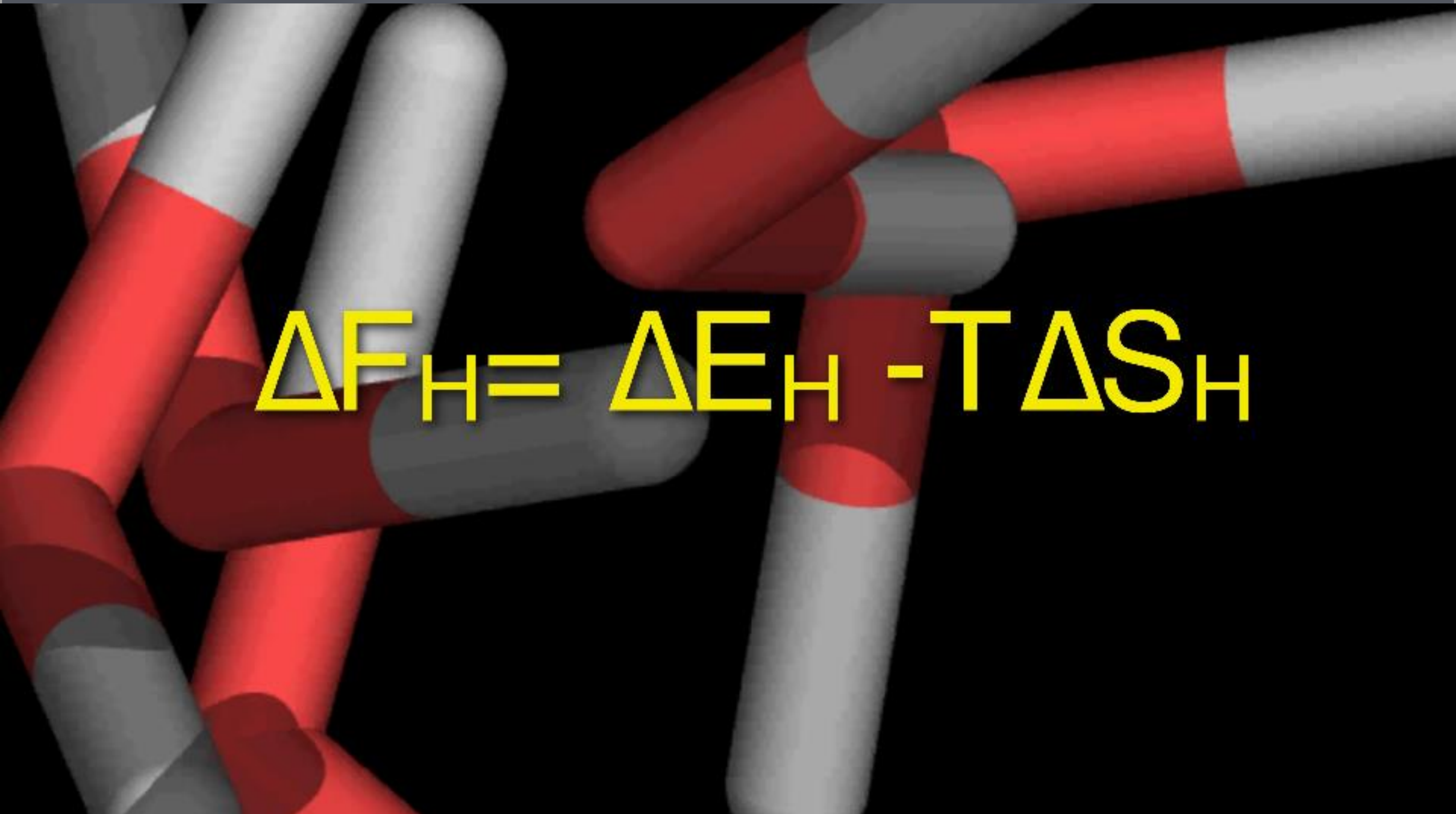
Loss: Entropy of 1 (0.5*2) freely rotating water ($\Delta S_H<0$)

Peer challenge

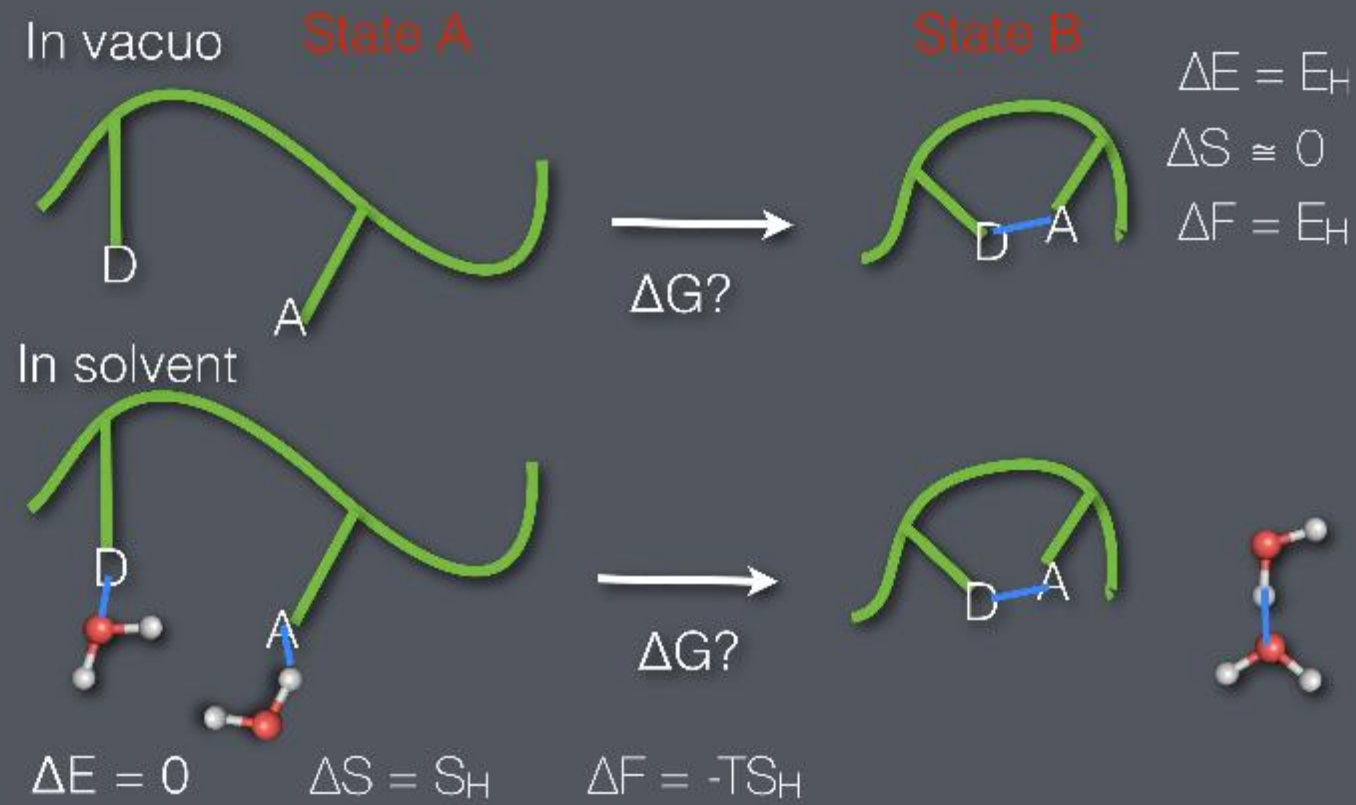
Which is true for H-bond formation?

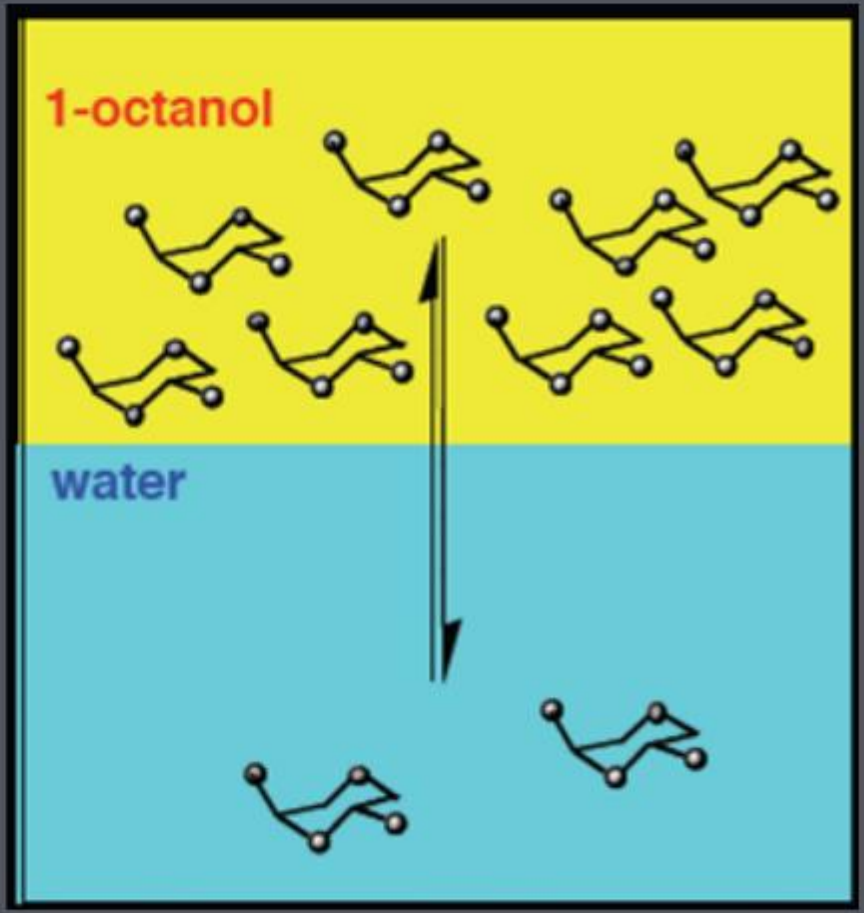
A) $\Delta E_H < T\Delta S_H$

B) $T\Delta S_H < \Delta E_H$


$\Delta F_H = \Delta E_H - T\Delta S_H$

H-bond ΔG for proteins





Why do some molecules
like oil/gas better?

Why do some molecules
like water better?

Partitioning - lab measurements!

Consider transfer of hydrocarbon to H_2O

Concentrations (X) rather than probability

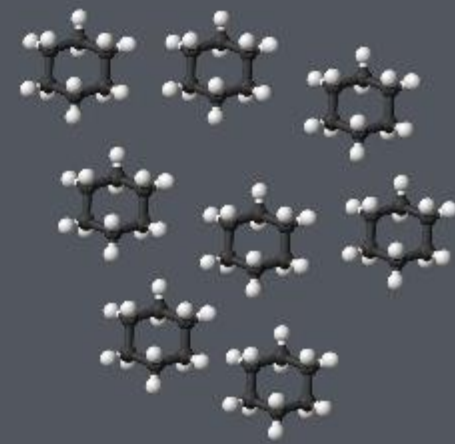
Count per mol, so we use R instead of k

$$X \propto \exp^{-\frac{G}{RT}}$$

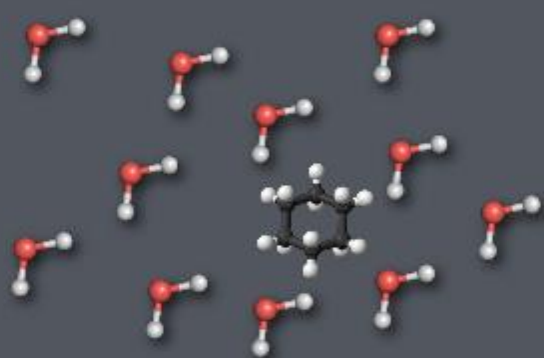
$$\Delta G_{\mathrm{liq}\rightarrow\mathrm{aq}} = -RT\ln\left(X_{\mathrm{aq}}/X_{\mathrm{liq}}\right)$$

Experimental

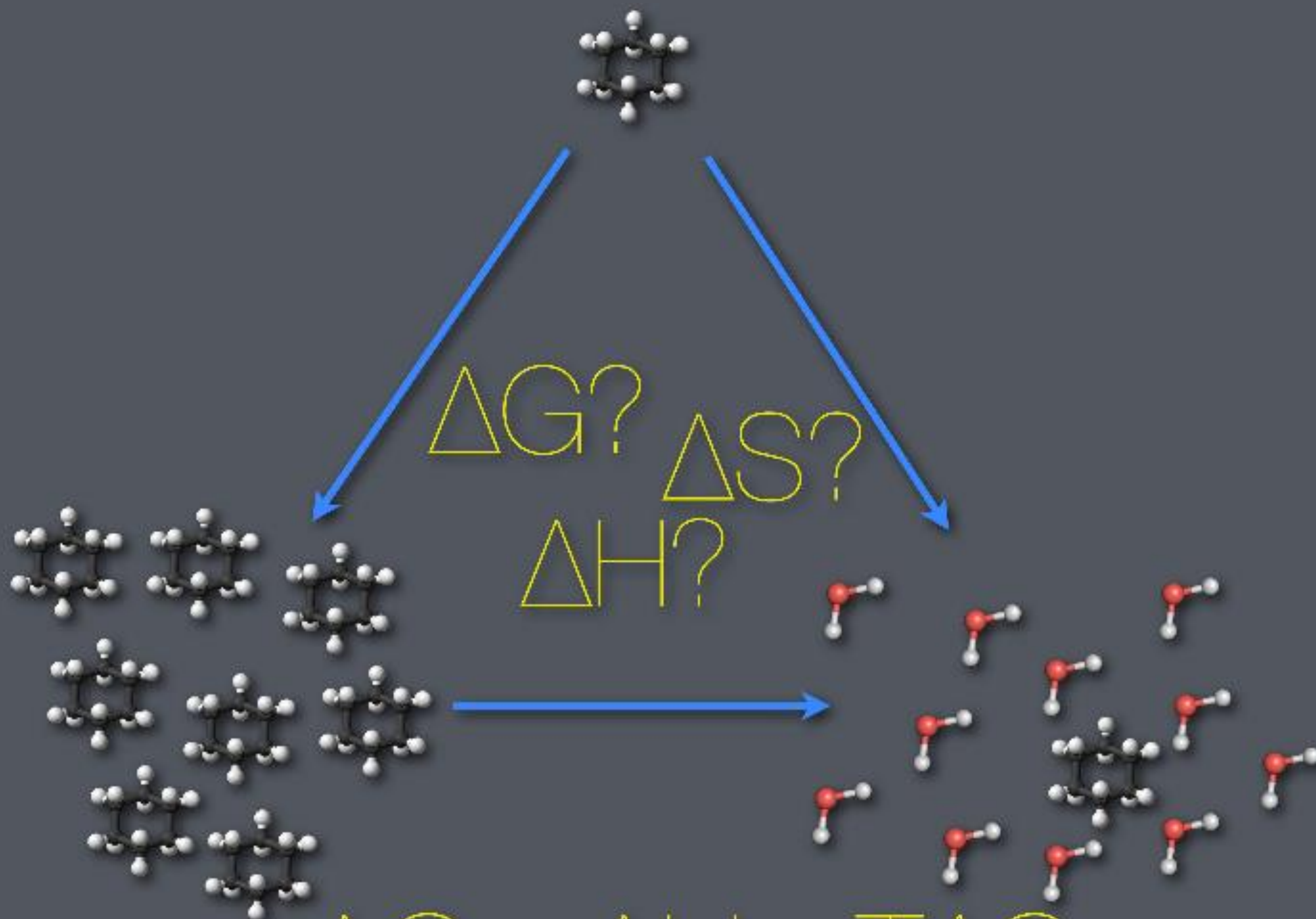
9.25 mol/l



$\Delta G_{liq \rightarrow aq} = +6.7 \mathrm{kcal/mol}$



0.0001 mol/l

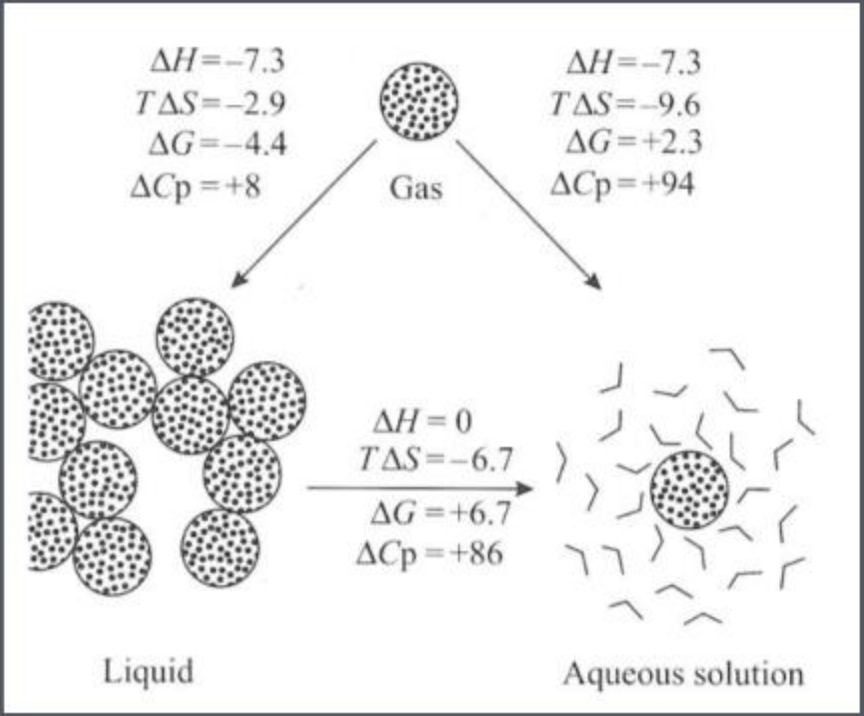


$$\Delta G = \Delta H - T\Delta S$$

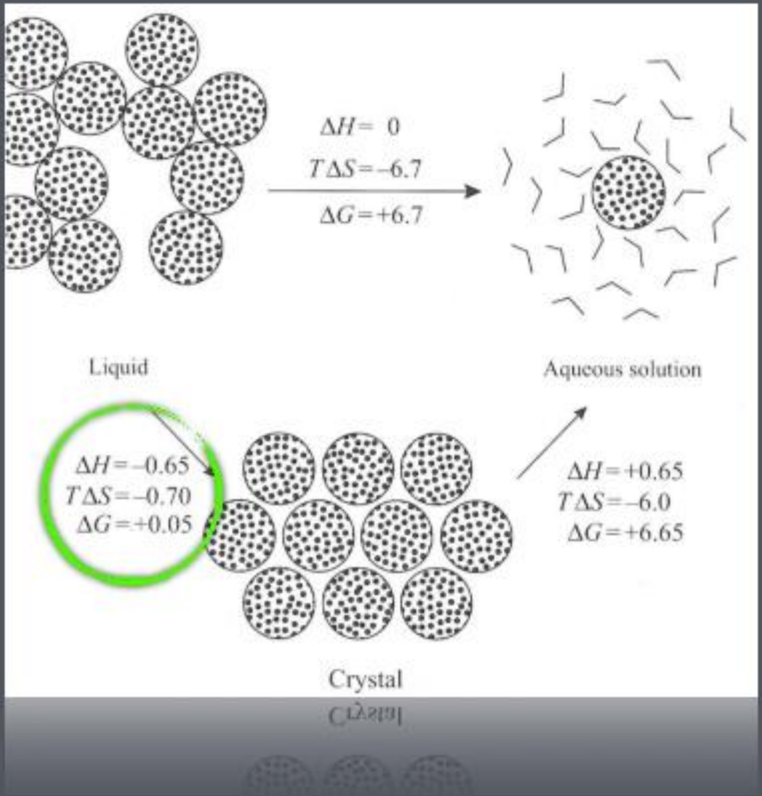
Hydrophobic solvation

We can compare the gas phase with aqueous or liquid phases the same way!

Knowing ΔG(T), we can calculate the other properties



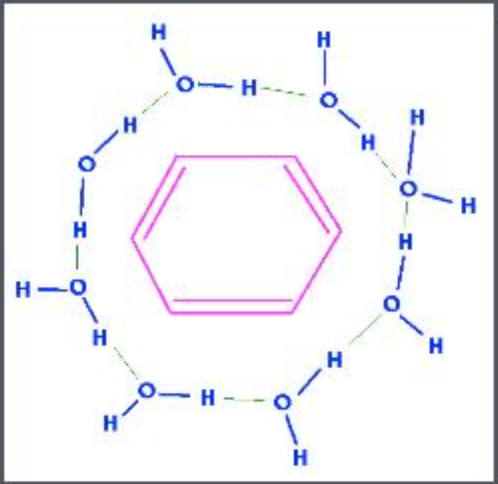
Hardening of structure

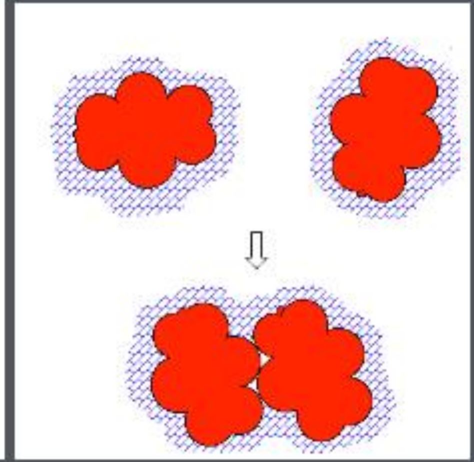


Once we have a separate hydrophobic phase, the cost is very low to "harden" it, or even form a crystal

What does this mean for proteins?

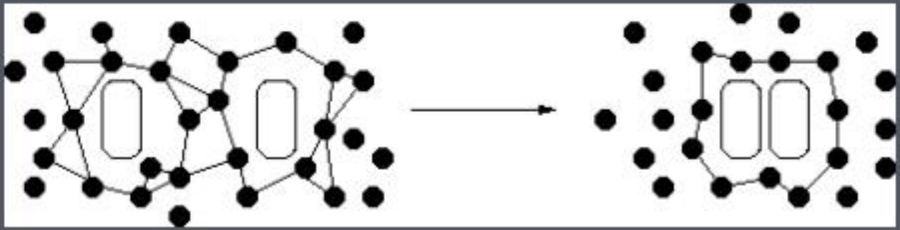
Hydrophobic effect





Can you account for these processes?

Clathrate structures



Study questions

1. What is an energy landscape?
2. How are energy landscape properties related to energy / entropy?
3. Explain the difference between states and microstates!
4. Based on what you now know, how would you explain entropy?
5. What is the hydrophobic effect? What interactions cause the hydrophobic effect?
6. For what can you use the Boltzmann distribution? Give concrete examples!
7. How does volume, or microstates, complicate the Boltzmann distribution?
8. What is the difference between energy and free energy?
9. Can you give examples of systems with (a) low, and (b) high entropy?
10. E*ntropy* is closely related to h-bonds and hydrophobic effect. How?
11. What is the difference between Helmholz & Gibbs free energy? (in theory & practice)
12. Give some examples of (a) relevant and (b) irrelevant energy barriers at 300K!
13. How can you predict what processes will occur? Have a look at https://bit.ly/2ARnMrj for a derivation!
14. What is the definition of temperature?
15. Why is hydrophobic ΔG so well correlated with nonpolar area?
16. What is the role of ε? Why is the water value so extreme?
17. What is ε roughly inside a protein?
18. What usually happens to titratable amino acids inside proteins?
19. Finkelstein says in one of the chapters "electrostatics in water originates not from energy, but from entropy". How can this be correct, given that hydrogen bonds are so important, and they are related to charge?

Covered material and suggested reading

- Nordlund
  - Chapter 14 (but better described in Finkelstein)
- Finkelstein
  - Chapter 4,5,6